

Estimating the Height of a Glider from its Shadow in an Aerial Orthophoto

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The glider over Isenau, Switzerland, in SWISSIMAGE (14 July 2023; 220 m $\times$ 52 m, north up). Imagery
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Abstract

A glider and its shadow both appear in the Swiss national orthophoto mosaic SWISSIMAGE. This note derives the glider's height above the ground from that single image and explains each step. The basic shadow formula gives 142.5 m, but it is biased. An orthophoto draws an airborne object where the camera ray through it meets the ground, not where the object is. We reconstruct the exposure geometry from public flight metadata: a pushbroom camera with a 45.3° field of view flying at 6393 m, 481 m to the side of the glider. The orthophoto therefore draws the glider about 20 m too far north, almost straight towards its own shadow. We also date the exposure to 11:25 UTC from the survey flight pattern and allow for the glider's own motion during the scan. The geometry then closes, and the estimate becomes

$$h = 180(3) \text{ m above the terrain beneath the glider} \quad (95\% \text{ interval } 173 \text{ m to } 186 \text{ m}).$$

Two independent clocks agree: the shadow direction and the flight log date the exposure to within 1.5 minutes of each other.

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1 The idea in one picture

Sunlight arrives as parallel rays at a known elevation angle α . The glider blocks one ray, and the shadow lands where that ray meets the hillside. If we know how far apart the glider and its shadow are horizontally (D), the height follows from a right-angled triangle, $D \tan \alpha$. A correction is needed because the hillside is not flat. That is all the shadow method is (Section 4).

The difficulty is that we never see the glider’s true horizontal position. The image is an *orthophoto*, rectified so that everything *on the ground* is at its map position. A glider 180 m above the ground is not on the ground. The camera saw it slightly from the side, and the rectification projects it along the camera’s line of sight onto the terrain (Figure 3). Here that slides the glider about 20 m towards its shadow, so the measured gap D is too short. Section 6 recovers the camera position and removes that error. This single correction moves the answer from 142 m to 184 m. The remaining steps are refinements of a few metres.

Step	h (m)	Section
Shadow formula, image positions taken at face value	142.5	4
+ camera relief displacement and terrain at the true position	184.1	6
+ glider motion during the pushbroom scan (30 m/s)	179.8	7
Monte Carlo over all input uncertainties (median $\pm 1\sigma$)	179.9 ± 3.3	10

Table 1: How the estimate is built up.

2 Data

Orthophoto. SWISSIMAGE is swisstopo’s national orthophoto mosaic. In this part of the Alps it is built from 0.25 m ground-sample-distance image strips. The covering strip is 20230714_1118_12504, flown on 14 July 2023. Its identifier encodes the start of acquisition, 11:18 UTC. Its footprint polygon, length (57.87 km) and start point are public ([data/imagery-acquisition.json](#)). The other eight strips flown that morning are also public ([data/flight-strips-2023-07-14.json](#)).

Camera. swisstopo flew a Leica ADS100 *pushbroom* sensor. It does not take frames. Each colour band is recorded by a single line of 20 000 pixels across the flight direction, and the image is built up line by line as the aircraft moves forward. The orthophoto uses the *nadir* line group, which looks straight down along-track. The camera’s SH120 head has a 120 mm lens and a nadir field of view of 45.2° across track. Its flying-height multiplier is 24 000:1, so 0.25 m pixels correspond to 6 000 m above ground.

Terrain. Elevations come from swissALTI3D, obtained via the geo.admin.ch height service. Local grids around the glider (2.5 m spacing) and the shadow (2 m) are cached in `data/terrain-grids.json` and bilinearly interpolated. The hillside under the glider falls about 30° to the east and gently to the north.

Frames. Horizontal coordinates are in CH1903+/LV95 (EPSG:2056). Combining grid scale and height reduction, LV95 coordinate differences equal ground distances to within 3×10^{-4} , i.e. 3 cm per 100 m. Only height *differences* enter the results, so the vertical datum cancels.

3 Measuring the image



Figure 1: Left: the glider, 40 m square, with its silhouette centroid (cross). Centre: the shadow area 76 m north-north-west, contrast-stretched. Right: the glider silhouette, blurred by the solar disk, matched to the shadow (red outline). The tail and fuselage land on the darkest blob, and the north-west wing lies along the dark band.

Both crops are fetched from the swisstopo WMS at 5 cm per pixel, upsampled from the 25 cm source (`analysis/measure_image.py`).

Glider. The glider is the only large white object, and thresholding brightness and saturation isolates its silhouette. The centroid is at $(E', N') = (2\,582\,425.79, 1\,135\,135.40)$, within 0.5 m of the original hand-placed pin. The silhouette spans 17.8 m, consistent with an 18-metre-class glider. The T-tail is to the north-east, so the glider was heading about 227° (south-west).

Shadow. The shadow is faint, and that is expected. The Sun’s disk subtends 9.3 mrad. From the glider–shadow distance along the ray, $(Z - z_s)/\sin \alpha \approx 220$ m, every edge is therefore smeared over about 2 m, more than twice the chord of the wing (about 0.7 m). No point on the ground is fully shaded by the wing, so its shadow is a half-tone smudge. The wider fuselage and tail make the darkest blob. To locate it objectively, we

1. turn the glider silhouette into a *predicted* shadow: invert it and convolve it with a disk of the penumbra width;

2. high-pass filter the shadow crop (subtract a 3 m Gaussian blur) to remove brightness gradients;
3. cross-correlate the two over all shifts, scales 0.85–1.15 and stretches along the sun direction 1.0–1.35, which allow for slope foreshortening.

The best match (Figure 1, right) has scale 0.97 and no stretch. Its peak stands 5.8 standard deviations above the background correlation. It is stable for penumbra widths of 1–2 m, and it puts the shadow’s centroid at $(E_s, N_s) = (2\,582\,416.12, 1\,135\,210.68)$. The correlation falls by one standard deviation within ± 0.8 m along the sun direction. Along the wing axis it falls more slowly, within ± 2 m. We carry $\sigma = 0.5$ m and 1.2 m along those two directions.

Because the shadow is a parallel projection of the glider, the centroid of the shadow is the shadow of the glider’s centroid. Matching centroids therefore pairs corresponding points.

Result. The image separation is $D' = 75.89$ m at azimuth 352.7° , with terrain $z_s = 1\,905.9$ m at the shadow and $z' = 1\,927.5$ m under the glider’s image.

4 Shadow geometry

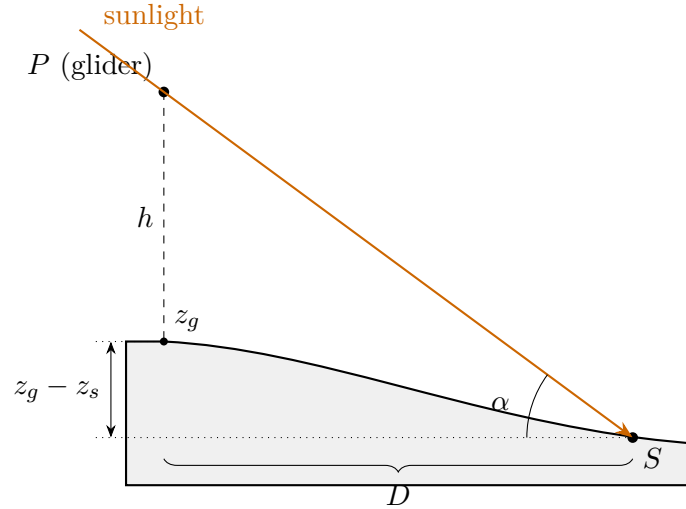


Figure 2: Vertical section along the sunlight. On a slope the shadow falls lower than the ground beneath the glider, and the terrain difference has to be removed.

Over a few hundred metres, sunlight is a family of parallel rays with elevation α and azimuth $A_\odot$. Earth curvature (1 mm over 100 m) and refraction (below $30''$ at $\alpha \approx 65^\circ$) are negligible. A glider point at height Z and horizontal position $\mathbf{g}$ casts its shadow at $\mathbf{s}$ on terrain of height z_s , with

$$Z - z_s = D \tan \alpha, \quad D = \|\mathbf{s} - \mathbf{g}\|, \quad \text{az}(\mathbf{s} - \mathbf{g}) = A_\odot + 180^\circ. \quad (1)$$

The height above the ground directly beneath the glider is $h = Z - z_g$, so

$$h = D \tan \alpha + z_s - z_g \quad (2)$$

Here the shadow falls about 22 m downhill, so the correction $z_s - z_g$ is negative. With the image positions taken at face value ($\mathbf{g} = (E', N')$) and $\alpha = 65.19^\circ$ (Section 5), Equation (2) gives $h = 75.89 \times 2.1625 + 1905.9 - 1927.5 = 142.5$ m. That is the *naïve* estimate. The next three sections correct its inputs.

Sensitivities. $\partial h/\partial D = \tan \alpha = 2.16$ and $\partial h/\partial \alpha = D \sec^2 \alpha = 7.5 \text{ m per degree}$. Under a high summer sun, horizontal errors are amplified more than twofold, and a degree of solar error costs several metres.

5 Exposure time and solar position

5.1 When was the glider photographed?

The strip ID gives only the start time. However, the nine strips flown that morning form a classic back-and-forth survey pattern, with alternate east- and westbound lines about 2.2 km apart. The interval between consecutive start times is the line’s length divided by the ground speed, plus a turn:

$$t_{k+1} - t_k = \frac{L_k}{V_{\text{dir}(k)}} + T_{\text{turn}}. \quad (3)$$

The start times are truncated to whole minutes. A least-squares fit with separate east- and westbound speeds, Monte-Carlo’d over the sub-minute truncation, gives the following (`analysis/fit_camera.py`):

- eastbound 398 km/h (95 %: 326–508);
- westbound 349 km/h (305–410);
- turn time 4.2 min.

The difference between the two speeds is consistent with a wind. The glider lies 44.13 km along the eastbound strip, which gives the exposure at

$$t = 11:18 + (7.15 \pm 0.91) \text{ min} = 11:25:09 \text{ UTC} \pm 55 \text{ s}.$$

5.2 Solar position

The Sun’s position follows Meeus’ low-precision algorithm (as in the NOAA solar calculator, accurate to about 1’), implemented in `paper/solar.mjs`. In outline, the Sun’s apparent longitude λ gives its declination, $\sin \delta = \sin \varepsilon \sin \lambda$. The time, the equation of time and the longitude give the hour angle $H_\odot$. With latitude φ ,

$$\sin \alpha = \sin \varphi \sin \delta + \cos \varphi \cos \delta \cos H_\odot, \quad (4)$$

$$A_\odot = \text{atan2}(\sin H_\odot, \cos H_\odot \sin \varphi - \tan \delta \cos \varphi) + 180^\circ. \quad (5)$$

At the glider (46.367 4° N, 7.210 3° E), $\delta = 21.68^\circ$ and local solar noon is at 11:37 UTC.

UTC, 2023-07-14	elevation α	azimuth $A_\odot$
11:18 (strip start)	65.01°	169.49°
11:25 (estimated exposure)	65.19°	173.41°
11:37 (solar noon)	65.31°	179.98°

Table 2: Solar position at the glider. An independent ASTROPY evaluation agrees to within 0.01°.

Close to solar noon the elevation hardly changes: a full minute of timing error moves α by only 0.02°, i.e. 0.2 m of height. The azimuth, in contrast, turns by 0.55° per minute. That makes the shadow direction a sensitive clock, which Section 9 uses as a check.

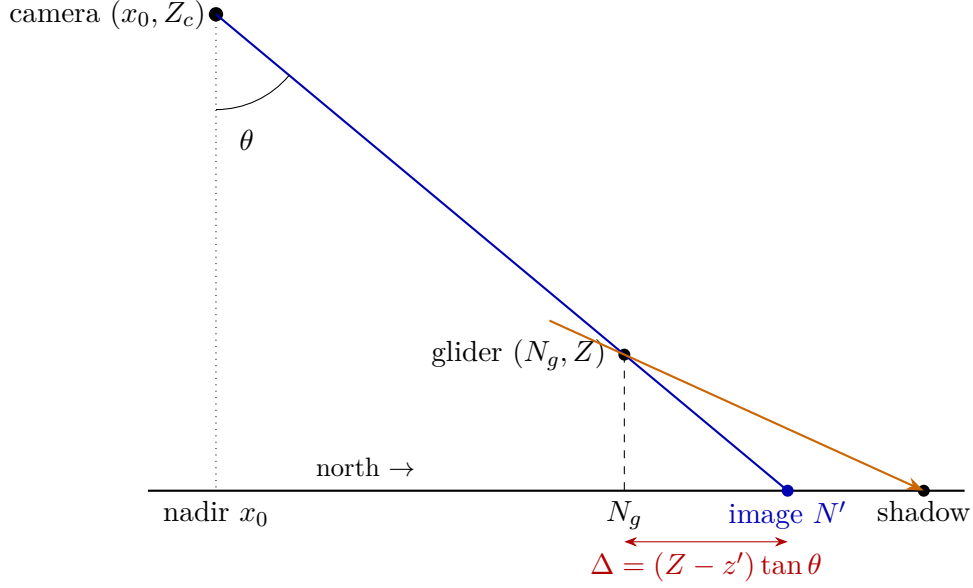


Figure 3: North–height section across the flight line (not to scale). The camera sees the glider at angle θ off vertical. The orthophoto puts it where that ray meets the ground, a distance Δ further from the nadir. Here the shadow also lies to the north, so the image of the glider moves towards its shadow and D is underestimated.

6 How an orthophoto draws a glider: relief displacement

6.1 Recovering the camera from the strip footprint

A pushbroom line looks out to $\pm\beta$ on either side of the nadir track. Over a ridge the swath edge meets the ground sooner, and over a valley later. A swath edge at terrain height z lies at horizontal distance $(Z_c - z) \tan \beta$ from the nadir track, so the footprint width across the track at any along-track position is

$$w = (Z_c - z_N) \tan \beta + (Z_c - z_S) \tan \beta = \tan \beta (2Z_c - z_N - z_S), \quad (6)$$

where z_N and z_S are the terrain heights at the north and south edges. This is linear in the unknowns $\tan \beta$ and $Z_c \tan \beta$. We measured w , z_N and z_S on 56 cross-sections of the footprint polygon, one per kilometre. The strip crosses Lake Geneva (372 m) and Alpine ridges above 3 000 m, which makes the fit well conditioned. The result is:

	fit (95 % bootstrap)	check
field of view 2β	45.30° (45.12°–45.49°)	datasheet SH120 nadir: 45.2°
flight altitude Z_c	6 393 m (6370–6417)	24 000:1 $\times$ 0.25 m = 6 000 m above $\approx$ 400 m
width residual	8.4 m RMS, $R^2 = 0.9997$	

The fitted field of view matches the manufacturer’s value to 0.1°, and the altitude matches the nominal pixel size over the lake. This confirms both the camera model and the constant-altitude assumption. Each cross-section then gives the nadir track as the average of the two edge solutions, $x_0 = \frac{1}{2}[(N_{\text{edge}} - (Z_c - z_N) \tan \beta) + (S_{\text{edge}} + (Z_c - z_S) \tan \beta)]$. The track is straight to 7 m RMS. At the glider it passes $N = x_0 = 1\,134\,654.6 \pm 1.3$ m, and we carry ± 4 m to cover local wander.

6.2 The correction

The glider's image lies $N' - x_0 = 481$ m north of the nadir track, and the camera is $Z_c - z' = 4466$ m above the ground there. The ray reaches the ground at $\tan \theta = 481/4466 = 0.1077$, i.e. $\theta = 6.2^\circ$ off vertical. By similar triangles, a point at height Z on that ray is drawn displaced away from the nadir by

$$\Delta = (Z - z') \tan \theta \approx 20 \text{ m} \quad (\text{northwards}), \quad (7)$$

with the true position $N_g = x_0 + (N' - x_0)(Z_c - Z)/(Z_c - z')$. Because the pushbroom's nadir lines look straight down along-track, there is no east-west displacement. The datasheet places the nadir group at 0° , and we allow $\pm 0.5^\circ$.

The sunlight runs 6.6° west of north, almost exactly along Δ . The true sun-ray distance is therefore $D = D' + (Z - z') \tan \theta_{\parallel}$, with $\tan \theta_{\parallel} = \tan \theta \cos 6.6^\circ = 0.1070$. Substituting into (1) gives an equation that contains Z on both sides. Solving it gives

$$Z - z' = \frac{D' \tan \alpha + z_s - z'}{1 - \tan \theta_{\parallel} \tan \alpha} = \frac{164.1 - 21.6}{1 - 0.1070 \times 2.1625} = \frac{142.5}{0.769} = 185.4 \text{ m}. \quad (8)$$

The denominator is the key quantity. Every metre the glider rises moves its image 0.107 m towards the shadow. That shortens the apparent baseline, and the shadow formula multiplies the shortfall by $\tan \alpha$, so the bias feeds on itself. Here the feedback factor is $1/0.769 = 1.30$: the naive estimate is 30% too low.

Finally, the ground must be read beneath the *true* position, 20 m south of the image, where the terrain is 1.3 m higher ($z_g = 1928.8$ m). Then $h = 184.1$ m.

7 Glider motion during the scan

A pushbroom image is not a snapshot. The eastbound aircraft (about 110 m/s) scans the shadow, which is 9.7 m further west, before the glider, so the two are recorded $\Delta t = 9.7/110 \approx 0.09$ s apart. The glider is flying at, say, 30 m/s on heading 227° . In that interval it moves 2.6 m south-west, away from its shadow. The shadow in the image was therefore cast from a point about 2.6 m north-east of where the glider is drawn, which is closer to the shadow. Allowing for this, with the general time offset

$$\Delta t = \frac{E_g - E_s + (Z - z_s) \tan \gamma}{V}, \quad (9)$$

where γ is the along-track look angle and V the aircraft ground speed, lowers the estimate to 179.8 m. The glider's airspeed is not measured. We take 20 m/s to 40 m/s (72–144 km/h), the normal range for a glider, and heading $227^\circ \pm 10^\circ$ to allow for wind drift. The white glider shows no colour fringes, which confirms that the colour bands of the nadir group were recorded simultaneously.

8 The full model

Collecting Sections 4–7, the unknowns are the glider's true position $\mathbf{g} = (E_g, N_g)$ and height Z at the instant the scan reached it. The model is

$$E_g = E' - (Z - z') \tan \gamma, \quad (\text{camera ray, along track}) \quad (10)$$

$$N_g = x_0 + (N' - x_0) \frac{Z_c - Z}{Z_c - z'}, \quad (\text{camera ray, across track}) \quad (11)$$

$$(\mathbf{s} - \mathbf{g} + \mathbf{v} \Delta t) \cdot \hat{\mathbf{u}} = \frac{Z - z_s}{\tan \alpha}, \quad (\text{sun ray}) \quad (12)$$

with $\hat{\mathbf{u}}$ the horizontal unit vector pointing away from the Sun and $\mathbf{v}$ the glider velocity. For given inputs this is one equation in Z , solved by root-bracketing. The height above ground is then $h = Z - z_{\text{DTM}}(\mathbf{g})$ (`analysis/solve_height.py`). The component of the sun-ray equation across $\hat{\mathbf{u}}$ is not used in the solution. It is kept as a residual that tests whether the geometry closes.

9 Consistency checks

1. **Field of view and altitude.** These come from the footprint alone and match the manufacturer’s datasheet (Section 6).
2. **Two clocks.** The shadow direction alone dates the exposure. After the relief correction, the shadow is exactly opposite the Sun at 11:26:35 UTC, which is 1.4 ± 0.9 min after the flight-log estimate of 11:25:09. Before the correction, the answer would have been 11:24. Both are well inside the survey window, and a shadow from another time of day would point tens of degrees away.
3. **Closure.** At 11:25 the cross-ray residual is 1.3 m for a stationary glider and -0.9 m for one moving at 30 m/s. Both are within the shadow’s positional uncertainty. The geometry closes without any free parameter tuned to make it do so.
4. **Shadow shape.** The matched template needs no stretch and a scale of 0.97. That is consistent with the glider image being enlarged across-track by $(Z_c - z')/(Z_c - Z) = 1.04$.

10 Uncertainty

10.1 Monte Carlo

All uncertain inputs are sampled together over 20 000 draws:

- exposure time $\mathcal{N}(7.15, 0.91)$ min after 11:18;
- glider centroid $\sigma = 0.15$ m; shadow centroid $\sigma = 1.2$ m along the wing axis and 0.5 m along the sun direction;
- $Z_c \mathcal{N}(6393, 12)$; $x_0 \mathcal{N}(\text{fit}, 4)$; $\gamma \mathcal{U}(\pm 0.5^\circ)$;
- glider speed $\mathcal{U}(20, 40)$ m/s; heading $\mathcal{N}(227, 10)$; aircraft $\mathcal{N}(110, 15)$ m/s;
- terrain: 0.5 m DTM error at each end, a grass offset of 0 to 0.5 m at the shadow, and 1 m uncertainty in the surface the orthophoto was projected onto.

Input group (varied alone)	σ_h (m)
Shadow and glider image positions	2.64
Glider motion during scan	1.49
Camera geometry (Z_c, x_0, γ)	0.92
Terrain	0.88
Exposure time / solar position	0.19
All together	3.3

Table 3: Error budget. The limiting factor is locating the faint shadow along the wing axis.

The result is

$$h = 179.9 \text{ m (median),} \quad 68\% : 176.5 \text{ m to } 183.1 \text{ m,} \quad 95\% : 173.3 \text{ m to } 186.2 \text{ m.}$$

10.2 Worst-case intervals (calculator)

The browser calculator in `glider-height/` implements Equation (2) without the camera model. It uses exact worst-case intervals instead of Monte Carlo. h increases with D , $\tan \alpha$ and z_s , and

decreases with z_g , so over an input box the extremes lie at corners:

$$h_{\min} = (D - \varepsilon_D) \tan \alpha_{\min} + (z_s - z_g) - 2\varepsilon_z, \quad (13)$$

$$h_{\max} = (D + \varepsilon_D) \tan \alpha_{\max} + (z_s - z_g) + 2\varepsilon_z. \quad (14)$$

To reproduce the corrected result (179.8 m) in the calculator, enter the position from which the glider cast the imaged shadow, i.e. corrected for relief displacement and for 30 m/s of motion, (2 582 427.72, 1 135 117.71), with $z_g = 1928.8$ m, the image-measured shadow (2 582 416.12, 1 135 210.68) with $z_s = 1905.9$ m, and $\alpha = 65.19^\circ$.

11 Alternative: reference masts

Instead of an ephemeris, α can be calibrated from a vertical mast of known height H , using $\tan \alpha = (H + z_b - z_t)/L$, where L is the mast's shadow length and z_b , z_t the terrain at its base and shadow tip. The calculator supports this, and two ski-lift masts nearby were estimated at about 9 m and 10 m (± 3 m) from ground photographs. At $\alpha \approx 65^\circ$, however, a 9 m mast casts only a 4.2 m shadow, and $\partial h / \partial H = D/L \approx 18$. A ± 3 m uncertainty in the mast height would therefore spread the glider height over ± 55 m. The ephemeris is about 300 times more precise here, so the masts are not used.

12 Limitations

- **Shadow identity.** The shadow is identified by position, orientation, size and shape (it matches the glider's silhouette at scale 0.97 and the correct wing angle). It is faint, however, and no statistical interval covers a misidentification.
- **Mosaic seams.** The orthophoto is assumed to come from strip 20230714_1118_12504. The strip covers the location, and the two clocks agree with it.
- **Glider attitude.** A banked glider has a different silhouette and shadow, but the matching is centroid-based and both silhouettes come from the same object.
- **Glider speed.** The speed is assumed, not measured. In principle it is encoded in the pushbroom distortion of the glider's image, which shears the wing-fuselage angle. Measuring that distortion could remove the second-largest error term.

Reproduction

<code>analysis/measure_image.py</code>	glider and shadow centroids from the WMS imagery
<code>analysis/fit_camera.py</code>	field of view, flight altitude, nadir track, exposure time
<code>analysis/solve_height.py</code>	full model, Monte Carlo, error budget
<code>paper/solar.mjs</code>	solar position
<code>glider-height/</code>	browser calculator and tested JavaScript module

Sources

- swisstopo SWISSIMAGE and its flight-strip metadata (layer `ch.swisstopo.lubis-bildstreifen`).
- swisstopo swissALTI3D terrain model, via the geo.admin.ch height service.
- Leica Geosystems, *Leica ADS100 product specifications*, 2016.
- J. Meeus, *Astronomical Algorithms*, 2nd ed., Willmann-Bell, 1998; NOAA solar calculator.